

Standard Situation Set-up

Ingredients:

- A. coffee powder/shot
- B. Cream/milk
- C. Water

Recipes:

Recipe #	Name	A	B	C	Mixing?
1	Americano	1	0	3	1
2	Latte	1	3	0	1
3	Light coffee	1	1	2	1
4	Coffee only	2	0	0	0
5	Milk only	0	2	0	0
6	Water only	0	0	2	0

Schematics Link: <https://a360.co/48bIKBP>

LCD

Workflow: LCD hovers on certain recipe and flashes until button 1 pressed to switch to next recipe, press button 2 to confirm, the recipe stops flashing and stay on.

Text Display

Status	Behavior	Text display	Text size (<16x2)
Hovering on 1	Constant on Flashing	SELECTING: 1-Americanano	10 11
Hovering on 2		SELECTING: 2-Latte	10 7
Hovering on 3		SELECTING: 3-Light Coffee	10 14
Hovering on 4		SELECTING: 4-Coffee	10 8
Hovering on 5		SELECTING: 5-Milk	10 6
Hovering on 6		SELECTING: 6-Water	10 7
Confirmed recipe choice	Constant on	COFIRMED	8

Canceled previous selection		CANCELED	8
Dispensing ingredients A/B/C		DISPENSING...	13?
During mixing		MIXING...	9?
After mixing		COMPLETE	8
No cup/removed during mixing/dispensing unsuccessful etc	Flashing	ERROR	5

Pseudo code for LCD:

1. Define recipes + ingredients:

Ingredient A= coffee

Ingredient B= milk

Ingredient C= water

Recipe 1: Americano

A = 2

B = 0

C = 1

mixing = yes

Recipe 2: Latte

A = 1

B = 1

C = 0

mixing = yes

Recipe 3: Light Coffee

A = 1

B = 0

C = 2

mixing = yes

Recipe 4: Coffee

A = 1

B = 0

C = 0
mixing = no

Recipe 5: Milk

A = 0
B = 1
C = 0
mixing = no

Recipe 6: Water

A = 0
B = 0
C = 1
mixing = no

2. Define states:

SELECTING
CONFIRMED
CANCELED
DISPENSING
MIXING
COMPLETE
ERROR

3. Selecting state:

If state = SELECTING:

LCD line 1 = "SELECTING:"
LCD line 2 = current recipe name

Make current recipe text flash

If button 1 is pressed:

move to next recipe
if current recipe index > 6:
go back to recipe 1

If button 2 is pressed:

state = CONFIRMED

4. Confirming logic:

If state = CONFIRMED:

LCD displays "CONFIRMED"

Wait for user decision:

If button 1 is pressed:

state = CANCELED

Else if button 2 is pressed:

check if cup exists

If no cup:

state = ERROR

Else:

state = DISPENSING

5. Cancellation logic:

If state = CANCELED:

LCD displays "CANCELED"

Wait short time

Return to SELECTING

Keep current recipe index unchanged

6. Dispensing logic:

If state = DISPENSING:

LCD displays "DISPENSING..."

Check for error:

If no cup

state = ERROR

Otherwise:

Read recipe and dispense

7. Mixing logic:

If state = MIXING:

LCD displays "MIXING..."

Check for error:

If no cup

state = ERROR

Otherwise:

mixing

Then:

state = COMPLETE

8. Complete logic:

If state = COMPLETE:

LCD displays "COMPLETE"

Wait

If button 1 is pressed

or button 2 is pressed:

return to SELECTING

9. Error logic:

If state = ERROR:

LCD displays "ERROR"

Make text flash

Turn off all dispensers

Turn off mixer

Wait until user acknowledges

If button 1 is pressed

or button 2 is pressed:

return to SELECTING

10. Init

When system powers on:

```
current recipe index = 1  
current state = SELECTING  
blink enabled = true
```

11. Main:

Repeat forever:

```
Read button 1  
Read button 2  
Check error conditions  
Update LCD based on current state
```

```
If current state is SELECTING:  
    handle hovering logic
```

```
Else if current state is CONFIRMED:  
    handle confirmed logic
```

```
Else if current state is CANCELED:  
    handle canceled logic
```

```
Else if current state is DISPENSING_A:  
    handle ingredient A dispensing
```

```
Else if current state is DISPENSING_B:  
    handle ingredient B dispensing
```

```
Else if current state is DISPENSING_C:  
    handle ingredient C dispensing
```

```
Else if current state is MIXING:  
    handle mixing logic
```

```
Else if current state is COMPLETE:  
    handle complete logic
```

Else if current state is ERROR:
 handle error logic

Pseudo code for Sensor

Function

Read force sensor value

If sensor value is greater than threshold

 cup exist = true

Else

 cup exist = false

 state = ERROR

If state = CONFIRMED

If cup exist = false

 state = ERROR

Else

 state = DISPENSING

If state = DISPENSING

 Keep checking sensor

If cup exist = false

 stop all pumps

 state = ERROR

If state = MIXING

 Keecp checking sensor

If cup exist = false

 state = ERROR

Component list & Spec sheet need to convert to a table

1. Motor: <https://www.pololu.com/product/2215>
2. Force Sensor: [Square Force-Sensitive Resistor \(FSR\) \[Alpha MF02A-N-221-A01\] : Adafruit Industries, Unique & fun DIY electronics and kits](#)
3. LED to demonstrate cup existence
4. LCD: [Standard LCD 16x2 + extras \[white on blue\] : Adafruit Industries, Unique & fun DIY electronics and kits](#)
5. Pump: